Weishu Zhang

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AI Infrastructure & LLM Serving

Education

University of California, Berkeley — Berkeley, CA
Bachelor of Science in Electrical Engineering and Computer Sciences — *Expected May 2027*

Technical Skills

Languages: Python, C/C++, TypeScript/JavaScript, SQL
Systems: Docker, Kubernetes, gVisor, PostgreSQL, Redis, REST APIs
AI / Tooling: PyTorch, vLLM, QLoRA, LLM APIs, pytest, OpenTelemetry

Experience

Amazon
Software Development Engineer Intern — *Sep 2026 – Present*
- Working on distributed training and inference infrastructure for AI Studio.

ASML — San Jose, CA
Software Engineering, Test Infrastructure — *May 2026 – Aug 2026*
- Built a daily regression-review workflow connecting Confluence, Jira, qTest, and compute jobs, consolidating test status, review context, and follow-up actions for QA.
- Structured the test-plan lifecycle into drafting, QA approval, asynchronous job execution, and result writeback, with explicit human checkpoints before execution.

Ludus — Berkeley, CA
Co-Founder & Software Engineer, Cloud Agent Infrastructure — *Apr 2026 – Aug 2026*
- Built a Kubernetes-based control plane for isolated agent workspaces, covering provisioning, suspend/resume, persistent workspaces, and TTL-based cleanup.
- Configured gVisor isolation, namespace-scoped RBAC, default-deny network policies, and resource quotas to bound sandbox permissions and resource use.

University of California, San Francisco — San Francisco, CA
Software Research Assistant — *Aug 2025 – Jan 2026*
- Developed a clinical-note extraction pipeline with schema-specific subtasks, output validation, and self-consistency checks to make long-context results easier to inspect.
- Built domain datasets and Qwen fine-tuning experiments with QLoRA; evaluated vLLM batching and quantization for inference throughput and memory use.

Open Source

PyTorch | *torch.compile, Dynamo* — Commit
- Landed a CELU validation fix aligning torch.compile with eager-mode behavior for alpha=0; added regression tests for numerical parity, in-place mutation, and invalid inputs.

Selected Projects

mini-vLLM | *LLM Serving Systems Simulator* — Code
- Implemented paged KV-cache allocation, reference-counted prefix reuse, copy-on-write, continuous batching, and recompute/swap preemption in a CPU-only simulator.
- Separated scheduling from a deterministic compute-cost model; added workload benchmarks and an interactive trace viewer for token budgets, cache occupancy, and preemption.
- Reproducible capacity experiment: paged allocation fits 112 sequences versus 16 for maximum-length reservation under the same 32,768-slot KV budget; compute is simulated.

GradeView / GradeSync | *Berkeley Course Infrastructure* — Live
- Refactored course-data services around PostgreSQL as the source of truth, with React, Node.js, and Python services plus Redis caching for repeated analytics queries.